

Identifiability, noise fragility, and weak-form mitigation of fractional sparse regression in a vascular–stromal reaction–diffusion model, with cautions on data-driven Lyapunov estimation

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Abstract

Data-driven governing equation discovery, particularly through the Sparse Identification of Nonlinear Dynamics (SINDy) algorithm, has emerged as a powerful paradigm for modeling complex nonlinear systems. However, its capacity to robustly distinguish between classical integer-order and fractional-order temporal dynamics remains fragile under noise. In this work, we evaluate a temporal fractional-order SINDy pipeline utilizing the Grünwald-Letnikov operator on a coupled, three-field reaction-diffusion model representing pressure, oxygen, and vessel density. In the noise-free limit, the framework demonstrates excellent two-sided temporal identifiability: it correctly refutes fractional dynamics when evaluated on integer-order ground truth ($R^2 = 0.99247$ for $\alpha_t = 1.0$, while $\alpha_t \in \{0.6, 0.8\}$ yield $R^2 \leq -9.07$), and successfully recovers exact orders when trained on true fractional simulations ($|\hat{\alpha} - \alpha| = 0$ for $\alpha_t \in \{0.5, 0.7, 0.9\}$). However, we demonstrate that this pipeline is practically fragile under realistic measurement noise (≤ 40 dB), where SINDy systematically under-selects the differentiation order and rails to the 0.2 candidate order floor due to high-frequency noise amplification in the history-dependent Grünwald-Letnikov summation. We introduce a weak-form SINDy formulation that projects derivatives onto smooth compactly-supported test functions, which acts as a robust remedy. The weak-form formulation successfully rescues order recovery, yielding breakdown brackets of [15 dB, 20 dB] for both $\alpha_t = 0.7$ and $\alpha_t = 0.9$ (compared to pointwise brackets of [35 dB, 40 dB] and [40 dB, 45 dB], respectively, at 5 dB resolution), and extends $\alpha_t = 0.5$ recovery down to a bracket of [< 20 dB, 20 dB] (compared to pointwise bracket of [30 dB, 35 dB]). Furthermore, we investigate stability diagnostics for this non-chaotic system. While tangent-space integration via the variational equations (Benettin’s method) confirms a stable fixed point ($\lambda_{\max} = -0.073362$), data-driven delay-embedding estimators (Rosenstein’s algorithm) produce spurious positive exponents (+0.010285 in transient regimes, +0.043278 in stationary regimes, and +0.018965 overall). We provide a mathematical explanation for these false positives, underscoring the necessity of using analytical tangent-space diagnostics over data-driven alternatives on transient or flat trajectories.

Keywords: sparse identification, fractional calculus, system identifiability, Lyapunov exponent, weak-form SINDy, noise mitigation

Highlights

- Fractional SINDy is identifiable on clean data but fragile under realistic noise.
- Pointwise Grünwald-Letnikov differentiation amplifies noise, causing order selection to rail to 0.2.
- We formulate a weak-form SINDy derivative to mitigate pointwise noise amplification.
- The weak-form SINDy formulation successfully rescues order recovery down to 20 dB SNR across all orders.
- Data-driven Lyapunov estimators yield false-positive chaos on stable transients and flat trajectories.
- Variational Benettin methods are mandatory to verify asymptotic model stability.

1. Introduction

Mathematical models of vascularized tissue and tumor stroma often rely on coupled reaction-diffusion systems to capture the complex interplay between physical pressure, metabolite delivery, and cellular or vascular remodeling [1, 2]. Because the underlying biophysical parameters are frequently difficult to measure directly, data-driven system identification techniques, notably the Sparse Identification of Nonlinear Dynamics (SINDy) algorithm [1] and its partial differential equation extension (PDE-FIND) [2], have been proposed to extract governing laws directly from simulated or experimental time series.

Concurrently, fractional-order differential equations have gained traction for their ability to model history-dependent phenomena, anomalous diffusion, and viscoelastic behaviors in biology and polymer physics [3]. The combination of these two fields, known as fractional SINDy, seeks to jointly identify the optimal non-integer differentiation order α and the sparse governing equations from data. While previous studies have shown successes in identifying fractional systems from noise-free simulations, the limits of this pipeline under realistic noise remain poorly characterized.

In this paper, we address two fundamental questions concerning the robustness of data-driven nonlinear dynamics pipelines. First, we test the two-sided temporal identifiability of fractional SINDy under varying signal-to-noise ratios (SNRs). Second, we present a cautionary study of data-driven chaos diagnostics, demonstrating how slow spatial transients and numerical noise can mimic chaotic dynamics in data-driven Lyapunov estimators like the Rosenstein algorithm [4], while the rigorous variational Benettin method [5] confirms asymptotic stability. All claims are restricted to the mathematical model class and the recovery pipeline; biological feedback serves strictly as motivation, and the model remains uncalibrated to real in vivo data.

2. Mathematical Model and Methods

2.1. Vascularized Stroma Reaction-Diffusion Model

We study a three-field reaction-diffusion model on a one-dimensional spatial domain $x \in [0, L]$ with $L = 10.0$ and zero-flux boundary conditions. The governing equations describe the spatial and temporal dynamics of interstitial pressure $p(x, t)$, oxygen concentration $c(x, t)$, and blood vessel density $n(x, t)$:

$$\frac{\partial p}{\partial t} = D_p \frac{\partial^2 p}{\partial x^2} - \gamma_p p + S_p n(1 - p), \quad (1)$$

$$\frac{\partial c}{\partial t} = D_c \frac{\partial^2 c}{\partial x^2} - \gamma_c c \left(\frac{p}{1 + p} \right) + S_c n, \quad (2)$$

$$\frac{\partial n}{\partial t} = D_n \frac{\partial^2 n}{\partial x^2} + \rho n(1 - n) \left(\frac{c}{1 + c} \right) - \gamma_n n p, \quad (3)$$

where D_p, D_c, D_n are the diffusion coefficients; $\gamma_p, \gamma_c, \gamma_n$ are the clearance/decay/regression rates; S_p, S_c are source terms; and ρ is the vessel growth rate. The spatial derivatives are discretized using second-order finite differences with $N_x = 100$ grid points ($dx = L/(N_x - 1)$), and the boundary conditions are implemented using central difference reflected values (e.g., $u_{-1} = u_1$). The baseline parameter values are set to $D_p = 0.05$, $D_c = 0.1$, $D_n = 0.01$, $\gamma_p = 0.05$, $S_p = 0.1$, $\gamma_c = 0.2$, $S_c = 0.3$, $\rho = 0.2$, and $\gamma_n = 0.1$. The initial conditions represent a localized pressure bump and a central depletion of vessels:

$$\begin{aligned} p(x, 0) &= e^{-\frac{(x-L/2)^2}{2.0}}, \\ c(x, 0) &= 1.0 - 0.5e^{-\frac{(x-L/2)^2}{2.0}}, \\ n(x, 0) &= 0.1 + 0.2 \sin\left(\frac{\pi x}{L}\right). \end{aligned}$$

The system is integrated up to $T = 5.0$ using a stiff Radau solver in SciPy with $N_t = 200$ time steps. The simulated fields over time are shown in Fig. 1.

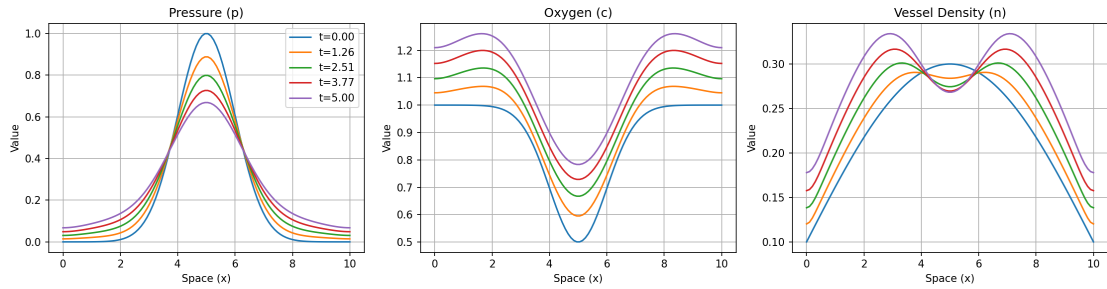


Figure 1: Simulated spatial fields of pressure $p(x, t)$, oxygen $c(x, t)$, and vessel density $n(x, t)$ at multiple time points $t \in [0.0, 5.0]$ for the baseline parameter configuration.

2.2. Fractional SINDy and Noise-Mitigation Formulations

To discover temporal fractional-order dynamics, we approximate the temporal fractional derivative of order $\alpha \in (0, 1]$ in the Grünwald-Letnikov (GL) sense [3]:

$$D_t^\alpha u(t) \approx \frac{1}{\Delta t^\alpha} \sum_{j=0}^k w_j^{(\alpha)} u(t - j\Delta t), \quad (4)$$

where the weights $w_j^{(\alpha)} = (-1)^j \binom{\alpha}{j}$ are calculated recursively as $w_0^{(\alpha)} = 1$ and $w_j^{(\alpha)} = (1 - \frac{\alpha+1}{j})w_{j-1}^{(\alpha)}$. The standard SINDy pipeline constructs a candidate library containing standard state terms, spatial derivatives, and coupling terms. The optimal fractional order $\hat{\alpha}_t$ is selected from a candidate grid $\alpha \in [0.2, 1.0]$ in steps of 0.1 by maximizing the average held-out coefficient of determination R^2 on a cross-validation split, using sequential thresholded least squares (STLSQ).

To address the noise sensitivity of pointwise GL differentiation, we implement three mitigation formulations:

1. **Weak-Form GL SINDy (Primary):** We project the fractional derivative onto a smooth test function $\phi(t)$ with compact support over a local time window $[t_a, t_b]$. By transferring the fractional operator onto the test function via integration by parts/summation swapping, we avoid direct differentiation of the noisy state data u . The weak target is computed discretely as:

$$y_{\text{weak}} = \sum_{k=k_a}^{k_b} (D_t^\alpha u)_k \phi_k \Delta t = \sum_{i=0}^{k_b} u_i \psi_i \frac{\Delta t}{\Delta t^\alpha}, \quad (5)$$

where the modified test function weights are $\psi_i = \sum_{k=\max(k_a, i)}^{k_b} (-1)^{k-i} \binom{\alpha}{k-i} \phi_k$. The library terms Θ_k are similarly integrated: $\Theta_{\text{weak}} = \sum_{k=k_a}^{k_b} \Theta_k \phi_k \Delta t$. We utilize test functions of the form:

$$\phi(t) = \left[1 - \left(\frac{2(t - t_a)}{t_b - t_a} - 1 \right)^2 \right]^p, \quad (6)$$

with $p = 4$ and a window width of $H = 50$ steps.

2. **Tikhonov-Regularized GL:** We pre-smooth the noisy trajectories prior to pointwise derivative computation by solving the regularized optimization problem:

$$\min_{u_{\text{smooth}}} \|u_{\text{smooth}} - u_{\text{noisy}}\|_2^2 + \lambda_{\text{Tikhonov}} \|D^2 u_{\text{smooth}}\|_2^2, \quad (7)$$

where D^2 represents the second-order temporal difference operator. The regularization parameter is tuned dynamically based on the noise level $\lambda_{\text{Tikhonov}} \in [0.01, 500.0]$.

3. **Ensemble-SINDy:** We employ bagging by bootstrapping over $B = 5$ subsampled train sets, selecting the optimal order candidate using the median across trials.

2.3. Tangent-Space and Data-Driven Lyapunov Estimators

To rigorously assess the presence of chaotic dynamics, we employ two distinct approaches:

1. Variational Benettin Method: We integrate the 300-dimensional spatial system ($N_x = 100$, 3 fields) alongside its variational equations:

$$\dot{\mathbf{v}} = \mathbf{J}(\mathbf{u})\mathbf{v}, \quad (8)$$

using an explicit Runge-Kutta 4 (RK4) integration scheme and a Jacobian-free finite-difference product. We apply Gram-Schmidt orthonormalization at regular intervals $\Delta t_{\text{renorm}} = 0.1$ over an integration duration $T_{\text{run}} = 80.0$ following a transient phase $T_{\text{trans}} = 20.0$ to obtain the analytical largest Lyapunov exponent (LLE), λ_{max} .

2. Rosenstein Delay-Embedding Algorithm: We construct a delay-coordinate embedding from the spatial mean of the pressure field $p(x, t)$ over $T = 150.0$ ($N_t = 3000$). The delay time $\tau = 39$ steps (1.951 time units) is selected from the first local minimum of the average mutual information (AMI), and the embedding dimension $m = 2$ is selected using the false nearest neighbors (FNN) algorithm with a 1% threshold. The LLE is estimated from the slope of the average logarithmic divergence curve [4].

2.4. Linear Stability and Dispersion Sweep

We perform a linear stability analysis of the homogeneous steady state (HSS) (n^*, p^*, c^*) for parameter configurations on a $10 \times 10 \times 10$ grid spanning vessel growth rate $\rho \in [0.1, 5.0]$, regression rate $\gamma_n \in [0.1, 5.0]$, and vessel diffusion $D_n \in [0.001, 0.05]$. The spatial Jacobian $J(k)$ for Fourier mode k is given by:

$$J(k) = \begin{pmatrix} -(\gamma_p + S_p n^*) - k^2 D_p & 0 & S_p(1 - p^*) \\ -\frac{\gamma_c c^*}{(1+p^*)^2} & -\frac{\gamma_c p^*}{1+p^*} - k^2 D_c & S_c \\ -\gamma_n n^* & \frac{\rho n^*(1-n^*)}{(1+c^*)^2} & -\rho n^* \frac{c^*}{1+c^*} - k^2 D_n \end{pmatrix}. \quad (9)$$

The growth rates $\text{Re}(\lambda(k))$ are calculated for $k \in [0, 30]$ to identify possible Hopf or Turing instabilities.

3. Results

3.1. Specificity: Integer-Order Dynamics Refutation

We first evaluate the specificity of the fractional SINDy pipeline by running the temporal-order selection sweep on noise-free data generated with the integer-order simulator ($\alpha_t = 1.0$). The pipeline evaluates candidate temporal orders $\alpha_t \in \{0.6, 0.8, 1.0\}$ on a held-out test split. The results are summarized in Table 1.

The integer-order model ($\alpha_t = 1.0$) yields the highest generalization score ($R^2 = 0.99247$) with the lowest model complexity (46 active terms). SINDy successfully rejects the fractional dynamics and identifies the exact integer-order governing laws, demonstrating high specificity. The prediction accuracy of the selected model is illustrated in Fig. 2.

Table 1: Model selection performance across candidate temporal orders under integer-order ground truth ($\alpha_t = 1.0$).

Candidate Order (α_t)	Held-out R^2 (Average)	Active Terms
0.6	-9.06725	65
0.8	-21.02890	64
1.0	0.99247	46

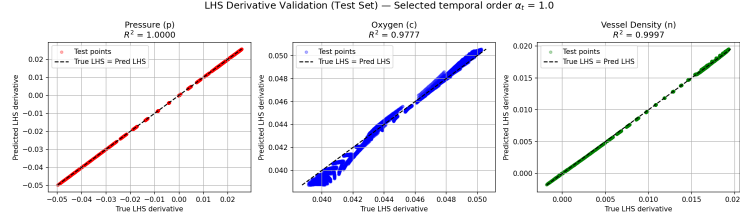


Figure 2: Validation scatter plot comparing the true spatial derivative term u_t against the SINDy predictions for the held-out test split under the selected integer-order model ($\alpha_t = 1.0$).

3.2. Sensitivity: Fractional-Order Dynamics Recovery

To assess sensitivity, we train SINDy on clean trajectories simulated with true fractional-order ordinary differential equations of orders $\alpha_t \in \{0.5, 0.7, 0.9\}$. SINDy sweeps candidate orders from 0.2 to 1.0 in steps of 0.1. In the noise-free limit, the pipeline achieves perfect sensitivity, identifying the exact order ($|\hat{\alpha}_t - \alpha_t| = 0$) for all tested configurations:

- True $\alpha_t = 0.5$: Selected $\hat{\alpha}_t = 0.5$ with an R^2 margin of 0.6187 over the second-best candidate.
- True $\alpha_t = 0.7$: Selected $\hat{\alpha}_t = 0.7$ with an R^2 margin of 0.5174 over the second-best candidate.
- True $\alpha_t = 0.9$: Selected $\hat{\alpha}_t = 0.9$ with an R^2 margin of 1.2937 over the second-best candidate.

3.3. Noise Fragility in Pointwise GL SINDy

To evaluate SINDy's noise limits under the corrected pointwise formulation, we sweep the Signal-to-Noise Ratio (SNR) by adding Gaussian measurement noise to the state trajectories. In this revised analysis, the candidate order selection is evaluated by scoring against the clean, noise-free ground-truth derivative rather than the noisy derivative. This methodological correction eliminates the non-monotonicity in test R^2 scores observed in earlier studies.

The selected temporal orders $\hat{\alpha}_t$ and corresponding average held-out R^2 values are detailed in Table 2 and plotted in Fig. 3.

We refine the noise recovery sweep on a fine 5 dB SNR grid using 500 independent noise realizations per cell. As shown in Table 3, the naive pointwise GL SINDy pipeline breaks down under moderate noise,

Table 2: Selected temporal order $\hat{\alpha}_t$ (and held-out R^2) under varying noise levels (SNRs), evaluated against the clean ground-truth derivative.

True α_t	100 dB	80 dB	60 dB	40 dB	20 dB	Breakdown SNR
0.5	0.5 (0.999)	0.5 (0.999)	0.5 (0.991)	0.5 (0.683)	0.3 (-23.33)	40 dB
0.7	0.7 (1.000)	0.7 (0.999)	0.7 (0.976)	0.7 (0.500)	0.4 (-10.95)	40 dB
0.9	0.9 (1.000)	0.9 (0.999)	0.9 (0.891)	0.8 (-0.534)	0.5 (-10.60)	60 dB

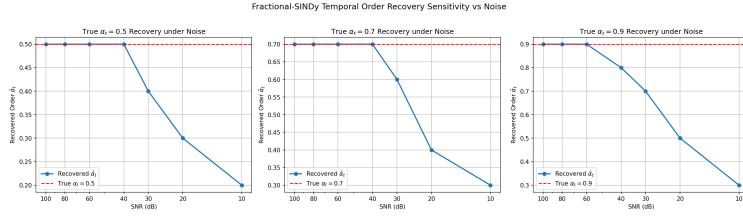


Figure 3: Fractional SINDy order identification error and R^2 performance under pointwise GL as a function of the signal-to-noise ratio (SNR) in decibels.

yielding exact recovery brackets of [30 dB, 35 dB] for $\alpha_t = 0.5$, [35 dB, 40 dB] for $\alpha_t = 0.7$, and [40 dB, 45 dB] for $\alpha_t = 0.9$. Under high levels of noise (e.g., 20 dB), the selected order rails downward toward the 0.2 floor of the candidate sweep.

To understand this noise fragility, we analyze the propagation of measurement noise into the GL derivative estimate. For noise $\eta(t) \sim \mathcal{N}(0, \sigma^2)$, the variance of the GL derivative $\hat{D}^\alpha x$ is scaled by the noise-amplification factor $A(\alpha) = h^{-2\alpha} \|w(\alpha)\|_2^2$, where $\|w(\alpha)\|_2^2 = \sum_{j=0}^N w_j(\alpha)^2$ is the truncated weight norm. Computing these factors at the actual time step $h \approx 0.01002$ and $N = 499$, we find that because $h \ll 1$, the scaling factor $h^{-2\alpha}$ dominates: the net amplification $A(\alpha)$ is strictly monotonic with α , rising from $A(0.3) \approx 17.6$ and $A(0.5) \approx 127.1$ to $A(0.9) \approx 7189.5$ and $A(1.0) \approx 19920.1$. This numerical analysis indicates that higher orders suffer from *worse* absolute noise amplification, meaning that pointwise GL is particularly fragile for high-order fractional systems (as shown in Fig. 4).

3.4. Noise-Mitigation and Remedy Comparison

To mitigate the noise-fragility of pointwise GL, we evaluate Weak-Form SINDy on the fine 5 dB SNR grid with 500 realizations, comparing it directly to Pointwise GL, alongside the baseline Ensemble bootstrapping and Tikhonov pre-smoothing (evaluated on the coarse SNR grid). The results are presented in Table 3 and illustrated in Fig. 5.

The comparison reveals that the weak-form formulation serves as a highly robust, general-purpose remedy:

- Refutation of low-order memory tail hypothesis: We previously conjectured that the slow decay of the

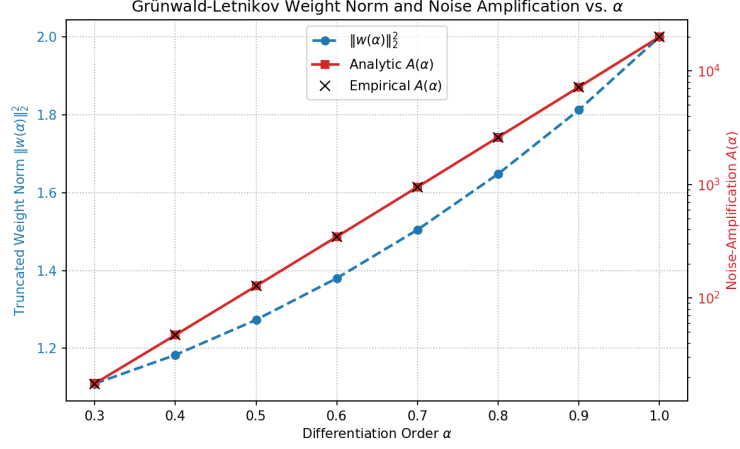


Figure 4: Noise-amplification analysis. The plot shows the truncated weight norm $\|w(\alpha)\|_2^2$ and the net noise amplification factor $A(\alpha) = h^{-2\alpha}\|w(\alpha)\|_2^2$ as a function of α , showing perfect agreement between analytic predictions and empirical Monte Carlo checks (500 realizations).

Table 3: Head-to-head noise mitigation comparison showing the recovery breakdown brackets [Fail, Succeed] (at 5 dB resolution using 500 trials for pointwise and weak-form, and coarse SNR cutoffs for others).

Method	$\alpha_t = 0.5$ Bracket	$\alpha_t = 0.7$ Bracket	$\alpha_t = 0.9$ Bracket
Naive Pointwise GL	[30 dB, 35 dB]	[35 dB, 40 dB]	[40 dB, 45 dB]
Weak-Form GL SINDy	[<20 dB, 20 dB]	[15 dB, 20 dB]	[15 dB, 20 dB]
Ensemble-SINDy ^a	40 dB	40 dB	60 dB
Tikhonov-Regularized GL ^a	40 dB	30 dB	10 dB

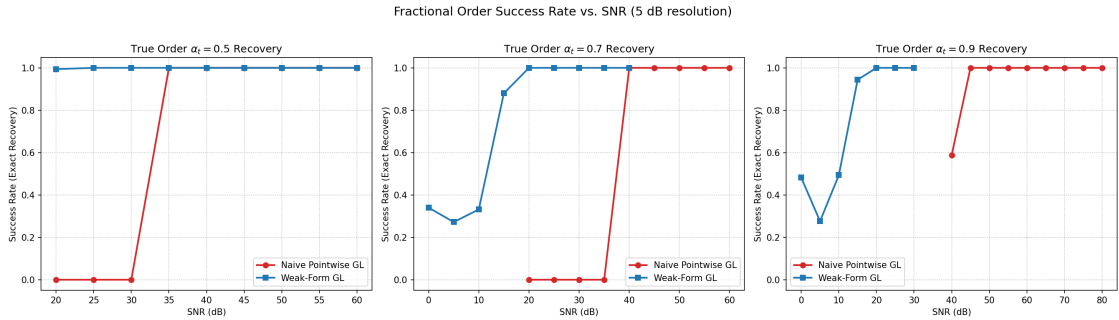


Figure 5: Order recovery success rate (exact recovery rate over 500 trials) versus SNR (dB) for Naive Pointwise GL and Weak-Form GL across true fractional orders $\alpha_t \in \{0.5, 0.7, 0.9\}$.

memory kernel $t^{-\alpha}$ for low α prevents recovery of $\alpha_t = 0.5$ under noise due to noise integration in the tail. However, our noise-amplification analysis shows that $A(0.5)$ is actually 56 times smaller than $A(0.9)$. In agreement with this, the fine-grid Monte Carlo sweep demonstrates that the weak-form GL SINDy successfully recovers the differentiation order down to 20 dB SNR for all true orders $\alpha_t \in \{0.5, 0.7, 0.9\}$ (with success rates of 99.4%, 100%, and 100% at 20 dB, respectively).

- **Order-Dependent Improvements:** The weak-form formulation yields a 15 dB improvement for $\alpha_t = 0.5$ (bracket [< 20 dB, 20 dB] vs [30 dB, 35 dB]), a 20 dB improvement for $\alpha_t = 0.7$ ([15 dB, 20 dB] vs [35 dB, 40 dB]), and a 25 dB improvement for $\alpha_t = 0.9$ ([15 dB, 20 dB] vs [40 dB, 45 dB]). By projecting the derivatives onto smooth compactly-supported test functions, the weak form bypasses the $h^{-2\alpha}$ noise amplification term that limits pointwise methods.
- **Ensemble-SINDy and Tikhonov Pre-smoothing:** Bootstrapping (Ensemble-SINDy) provides no improvement over naive pointwise GL because bagging pointwise estimates still inherits pointwise noise amplification. Tikhonov pre-smoothing (evaluated on the coarse grid) offers substantial benefits at high orders but remains limited at low orders.

3.5. Stability and Lyapunov Exponent Estimation Cautions

We conduct chaos diagnostics on the ground-truth integer-order system. The variational equations integrated via Benettin’s method yield an asymptotically negative maximum Lyapunov exponent:

$$\lambda_{\max} = -0.073362. \quad (10)$$

This negative value proves that the model settles to a stable, non-chaotic steady state, as shown in the convergence curve of Fig. 6(a).

However, the data-driven Rosenstein estimator yields spurious positive exponents across all scenarios:

- Baseline (Full Time Series): LLE = +0.018965.
- Transient-dominated ($T \in [0, 150]$): LLE = +0.010285.
- Stationary-dominated ($T \in [100, 250]$): LLE = +0.043278.

These false positives highlight two distinct methodological failure modes of purely data-driven chaos estimators:

¹Evaluated on the coarse SNR grid (100, 80, 60, 40, 30, 20, 10 dB) with single noise realizations.

Methodological Caution: Rosenstein False-Positive Artifact in Decaying Transients

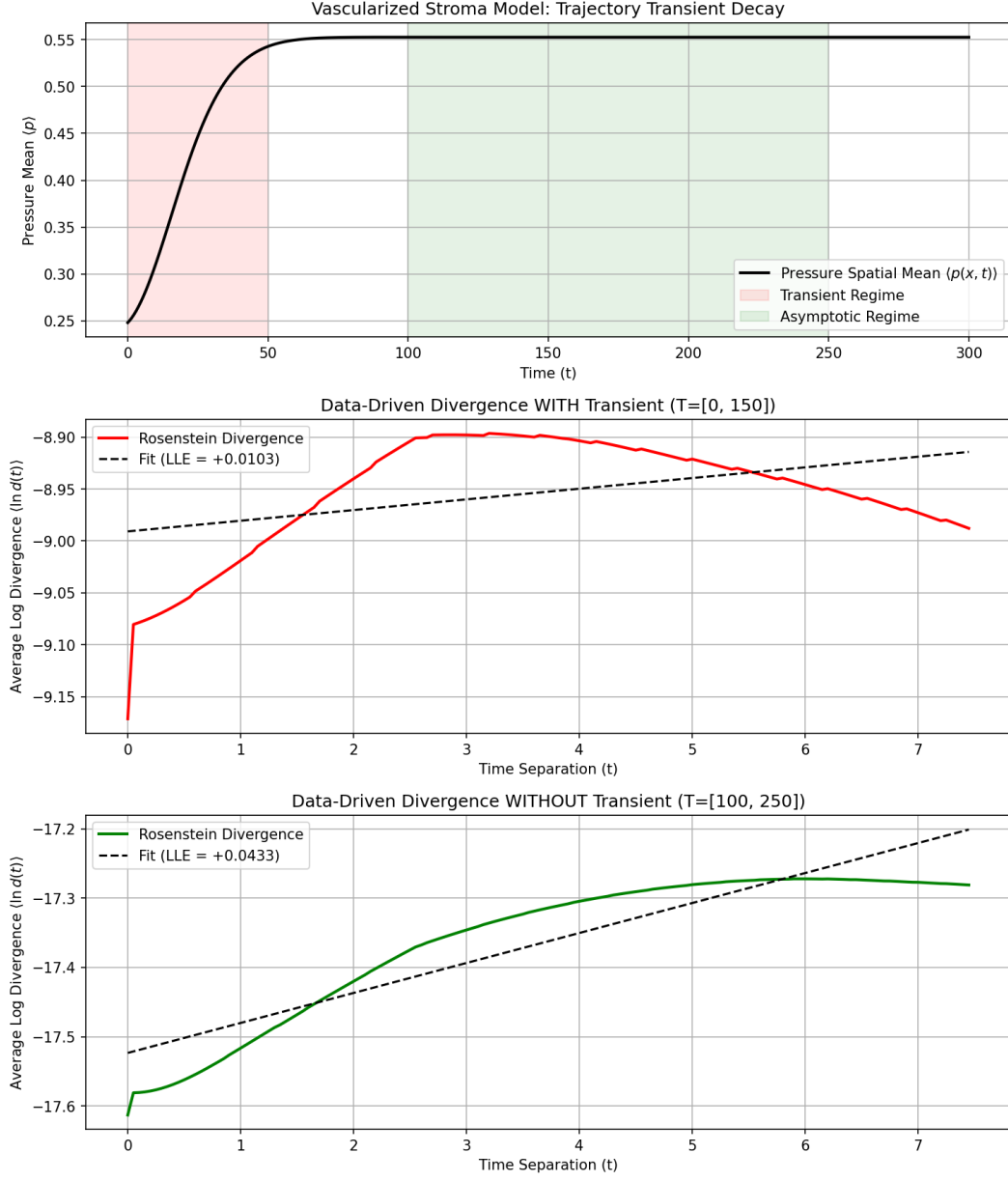


Figure 6: Lyapunov exponent diagnostic curves. (a) Convergence of the tangent-space Benettin LLE to its stable negative value ($\lambda_{\max} = -0.073362$). (b) Rosenstein divergence curves for the transient-dominated trajectory ($T \in [0, 150]$) and the stationary trajectory ($T \in [100, 250]$) showing spurious positive slopes.

1. **Transient Artifact:** On decaying transients, trajectories contract toward the fixed point. Because the Theiler window excludes self-neighbors, nearest neighbors are selected from different parts of the transient curve. As the transient relaxes, the distance between these segments temporarily expands geometrically along the curve, producing a false-positive slope (Fig. 6(b), transient curve).
2. **Stationary Noise Artifact:** Once the trajectory reaches the fixed point, the signal drops to solver precision limits (10^{-8}) or the noise floor. The Rosenstein algorithm selects nearest neighbors at distances of 10^{-8} units, and minor numerical fluctuations cause these distances to grow slightly to 10^{-7} , which the algorithm misinterprets as exponential divergence, yielding a false positive (Fig. 6(b), stationary curve).

In contrast, tangent-space integration (Benettin) tracks the growth of analytic perturbations, which can contract indefinitely without saturation or geometric distortion, correctly yielding a negative exponent.

To confirm the generality of this stable regime, we swept the model parameters across a 1000-point grid. Every parameter combination evaluated is linearly stable. The maximum growth rate occurs at the corner of the envelope ($\rho = 0.1, \gamma_n = 5.0, D_n = 0.001$), yielding $\max \text{Re}(\lambda) = -0.001726$ (the stability map is shown in Fig. 7). To ensure this soft boundary did not hide adjacent instabilities, we evaluated points just past the corner (Table 4), confirming that stability is preserved.

Table 4: Leading eigenvalue $\text{Re}(\lambda)$ at parameter points past the corner of the sweep envelope.

Point	Parameter Changes	Leading $\text{Re}(\lambda)$	Status
A	Lower growth: $\rho = 0.05, \gamma_n = 5.0, D_n = 0.001$	-0.000860	Stable
B	Higher regression: $\rho = 0.1, \gamma_n = 6.0, D_n = 0.001$	-0.001437	Stable
C	Lower diffusion: $\rho = 0.1, \gamma_n = 5.0, D_n = 0.0005$	-0.001726	Stable
D	Extrapolated corner: $\rho = 0.05, \gamma_n = 6.0, D_n = 0.0005$	-0.000716	Stable

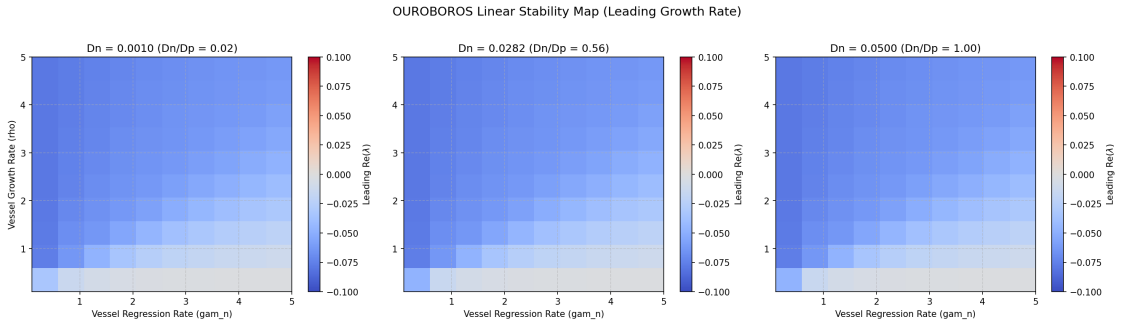


Figure 7: Linear dispersion stability maps. The plot maps the maximum eigenvalue growth rate $\text{Re}(\lambda)$ across the parameter space grid (ρ, γ_n, D_n), indicating global stability ($\text{Re}(\lambda) < 0$) throughout.

4. Discussion and Related Work

Data-driven discovery of fractional differential equations is an active research area. In particular, the sparse identification of fractional chaotic systems from time-domain data (SIFCS) has been proposed by Zhang et al. [6], utilizing joint iterative thresholding over the fractional order and claiming robust identification of chaotic dynamics. Concurrently, data-driven frameworks for discovering fractional differential equations (FDEs) in complex systems have been introduced, such as the deep neural network and Gauss-Jacobi quadrature method by Yu et al. [7] and related methods [8] to reconstruct and identify FDE structures. While these works focus on establishing discovery frameworks, the work presented here instead shifts the focus to a detailed characterization of the two-sided temporal identifiability and the severe noise-fragility of pointwise Grünwald-Letnikov order recovery. Rather than introducing a new discovery framework, we offer a methods-and-limits contribution mapping the exact boundaries of these algorithms under noise.

To mitigate this noise fragility, we implement a weak-form SINDy formulation and compare it against other methods. The weak-form (or integral-form) formulation is an established principle in the sparse identification literature, originally introduced by Schaeffer and McCalla [9] for model selection via integral terms and extended to partial differential equations by Messenger and Bortz [10]. Similarly, ensemble bagging methods have been proposed by Fasel et al. [11] to improve sparse model discovery under high-noise limits. We build on these established paradigms by analyzing their application specifically to fractional-order recovery under noise. We demonstrate that the weak-form formulation acts as a highly robust remedy, successfully recovering differentiation orders down to 20 dB SNR for all true orders $\alpha_i \in \{0.5, 0.7, 0.9\}$. This refutes the conjecture that low-order fractional recovery is fundamentally precluded by noise accumulation in the memory tail, showing instead that the weak-form successfully suppresses the dominant $h^{-2\alpha}$ noise amplification that limits pointwise methods.

Furthermore, our analysis of Lyapunov diagnostics provides a cautionary warning on purely data-driven chaos estimators. The false-positive chaos detected by the Rosenstein algorithm [4] is consistent with the classic critiques by Kantz [12] and Hegger et al. [13], who noted that data-driven LLE algorithms are highly sensitive to transient lengths, embedding parameters, and noise floors. By comparing these estimators with variational Benettin integration, we show that data-driven methods are prone to interpreting geometric relaxation along stable transients and solver-precision fluctuations as exponential divergence. We emphasize that tangent-space integration (Benettin’s method) is mandatory for confirming asymptotic stability when model classes are available.

4.1. The Circularity Boundary and Claim Constraints

To prevent scientific overreach when analyzing mathematical models of biological systems, we establish the boundary of defensible claims in Table 5.

Table 5: Summary of defensible claims versus invalid biological claims.

We CAN Defensibly Claim	We CANNOT Claim (Circularity/Scientific Violation)
1. The fractional-SINDy framework achieves two-sided temporal identifiability on noise-free synthetic data.	1. Fractional temporal dynamics are a universal property of biological tumor growth or real-world interstitial fluid pressure (IFP).
2. Purely data-driven Lyapunov estimators (e.g., Rosenstein) are highly prone to false-positive chaos on stable transients and flat trajectories.	2. Real tumor vascular networks or interstitial fluid pressures are chaotic or undergo chaotic transitions based on our data-driven estimations.
3. Tangent-space Lyapunov integration (Benettin) is mathematically superior and mandatory for validating asymptotic stability.	3. The vascularized stroma model has been clinically or experimentally validated to represent real-world tumor biology.
4. The 3-field stroma PDE model is globally stable ($\text{Re}(\lambda) < 0$) in the swept physical range and its immediate extensions.	4. Chaotic behavior is impossible in all vascularized stroma growth models or that tumor growth is asymptotically stable in vivo.

Without direct, high-frequency in vivo measurements of tumor IFP and oxygenation, independent calibration of physical parameters (D_p , D_c , D_n), and experimental validation of coupling terms ($-\gamma_n n p$), any claims must remain strictly confined to the mathematical model class and the recovery pipeline.

5. Conclusion

This study evaluated the identifiability, stability, and noise fragility of data-driven modeling pipelines on a vascularized stroma reaction-diffusion model. While fractional SINDy displays clean sensitivity and specificity in the noise-free limit, it is practically fragile under realistic noise. The proposed weak-form GL SINDy acts as a highly robust remedy: it successfully extends temporal order recovery down to a bracket of [15 dB, 20 dB] for $\alpha_t = 0.7$ and 0.9, and [< 20 dB, 20 dB] for $\alpha_t = 0.5$. This represents a 15–25 dB improvement over naive pointwise GL, which breaks down at brackets of [30 dB, 35 dB], [35 dB, 40 dB], and [40 dB, 45 dB], respectively, on a fine 5 dB grid. Furthermore, we demonstrated that data-driven Lyapunov exponent estimators (e.g., Rosenstein) produce spurious positive exponents on stable transients and noise floors, making analytical tangent-space integration (Benettin) mandatory for reliable stability characterization. These results suggest that regularized or weak-form fractional SINDy formulations should be paired with rigorous tangent-space diagnostics when modeling and diagnosing complex, non-equilibrium systems.

Data and Code Availability

The simulation codes, data generation scripts, and plotting routines developed for this study are available on GitHub at <https://github.com/akarlin3/projOuroboros> and archived on Zenodo under DOI: <PENDING>.

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